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Physics 327 Section 1

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LAB 9: LABVIEW and GPIB

Introduction

This lab is designed to practice using computer-based control, automation, and data acquisition systems by using a GPIB device which can export gathered data to the LabView platform. LabView features a front panel and a block diagram, where the front panel displays the simplified interface running a certain procedure defined by the virtual instrument, constructed in the Block Diagram in a way similar to programming. It gives a more clear view of the input and output of the VI. Compared to code-based programming languages, it's easier for beginners and easier for anyone to use a written VI file with such interface separation feature, and block programming can reduce time debugging as the VI reports errors clearly, while constructing the diagram would be more troubling when involving more complex variables, as each variable and function needs to be located in a menu.

Experiment Setup

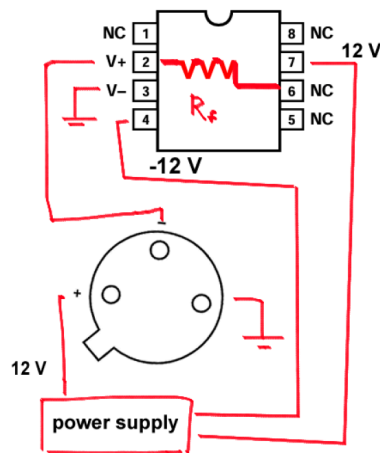
The example virtual instrument is first used to familiarize using LabView. [Question 1] The black box of the front panel seems to contain a low pass filter, where output filters the signals by a certain threshold of frequency.

After testing several LabView features, GPIB, connected to the computer, is powered on with the GPIB Write feature turned on in LabView to measure and write DC voltage output from a circuit. [Question 2] The HP multimeter is to be set up using the command `conf:volt:dc`

to switch to DC voltage measurement feature. Similarly, it's also configured to measure the resistance. A repeating measurement of the resistance of a thermistor at room temperature is initially done, then it's repeated with a hand holding the sensor, which means its temperature should increase, hence the resistance should decrease.

A circuit powering on the AD590 temperature sensor is constructed as Figure 1, where $R_f = 10\text{ k}\Omega$. In this schematics, LM741 converts the output current from AD590 to a stable and readable voltage through the feedback resistor and outputs such voltage through pin 6.

Figure 1

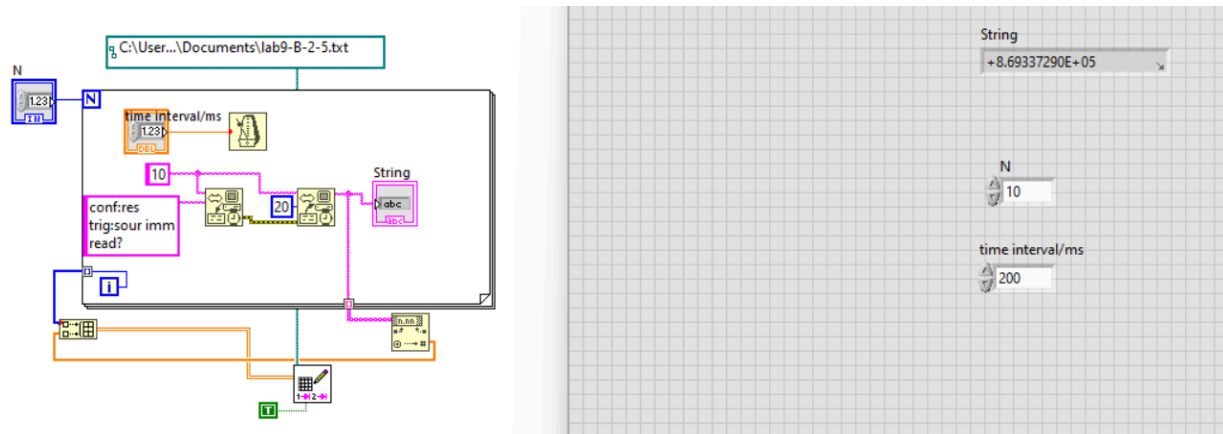


The measurement is switched to read a DC voltage. The VI is to be modified so that an LED in the front panel would light up signaling the temperature exceeding 300K by adding boolean features. Throughout the lab, the time interval was taken to be Results are to be recorded and analyzed.

Results and Discussion

Figure 2 shows the schematics of the measurement of resistance of the thermistor, with an interval of 1 second. [Question 3] The thermistor's value is initially around $8.63 \cdot 10^5\ \Omega$.

Figure 2



Where the string shows the measurement. Table 1 shows the measurement of thermistor resistance at room temperature.

Table 1

time/s	resistance/Ohm
0	863023.31
1	862961.49
2	862917.96
3	862882.93
4	862856.66
5	862834
6	862807.73
7	862791.24
8	862779.39

9	862763.42
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Holding the thermistor, Figure 3 shows the plot of the resistance with 10 measurements with the interval of 1 second. Figure 4 shows the plot after releasing it with 20 measurements, as a few seconds were left for confirming the program started before releasing.

Figure 3

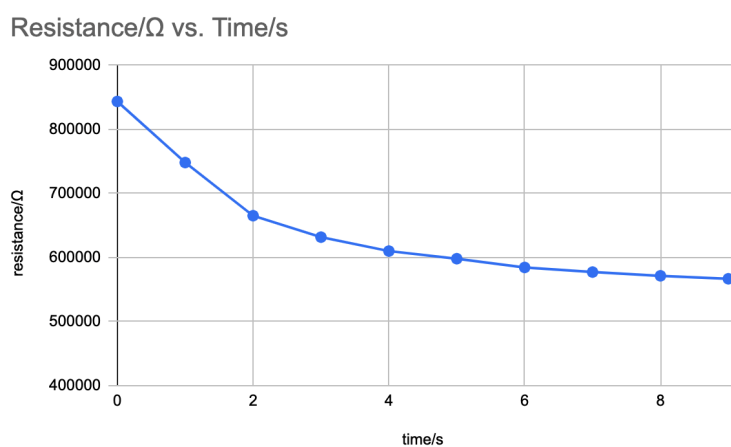
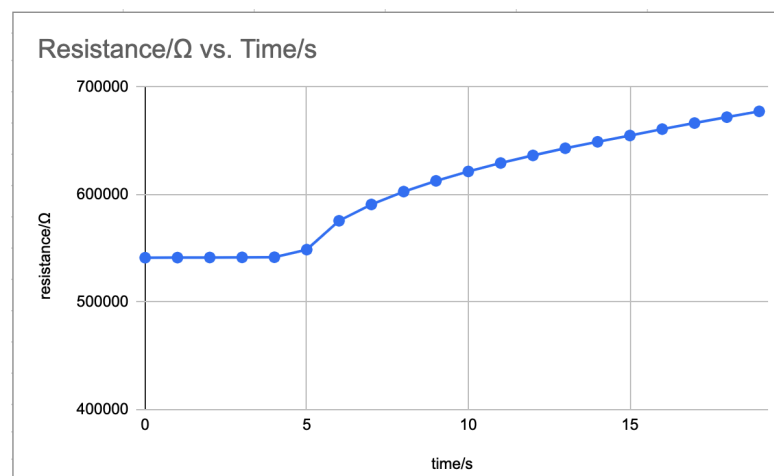


Figure 4



When holding the thermistor, the thermistor's resistance decreased in an exponential pattern as the time elapsed; after releasing it, it increased exponentially. Both meet the expectation such that the temperature's growth follows an exponential pattern.

Assembling an op-amp circuit with AD590, tapping and releasing the temperature transducer, Figure 5 shows the original figure outputting temperature vs. time, where the temperature conversion was done wrong by a factor of 3. DC voltage output should be multiplied by -100 , in this case it's mistaken to be -300 . 30 measurements with a timer interval of 1 second were taken. Figure 6 shows the corrected version of another trial's data.

Figure 5

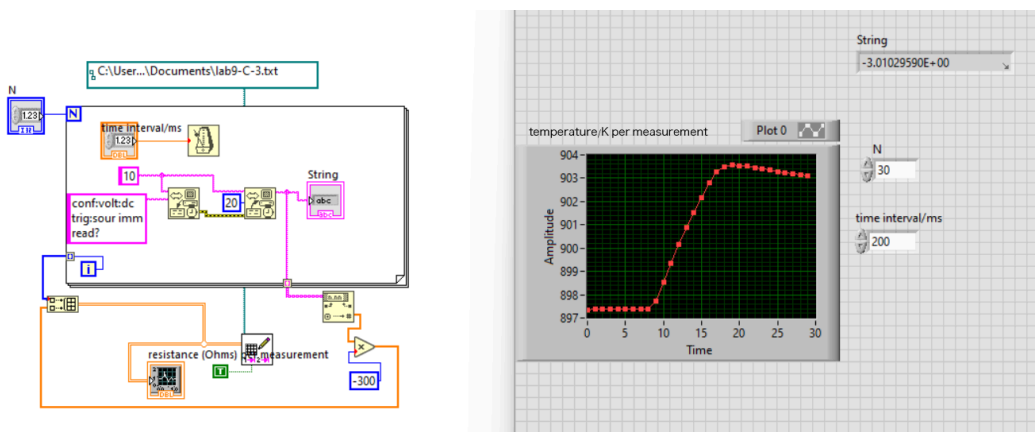
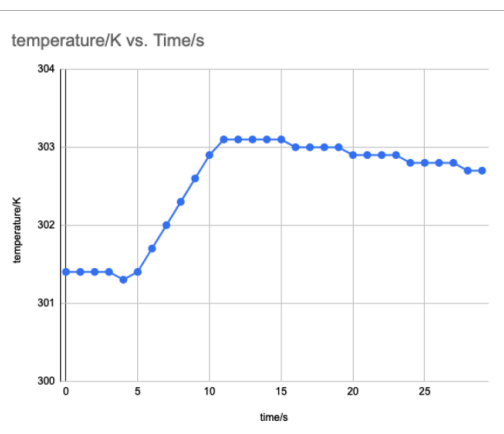


Figure 6



[Question 4] The output voltages are all around -3.0 V , giving the room temperature to be around 300 K , which is a reasonable value. As expected, it initially increased exponentially when tapping it, then decreased slowly, as featured by AD590. As the time ran out, the boolean comparison feature wasn't launched. The VI up with displaying temperature graph. Though a pattern can't be observed from the later plotted graph, from the initial graph in Figure 5, it could be observed that there's a vague exponential decreasing pattern. To better fit an exponential regression, heating the sensor more with a longer time and measurement interval could possibly improve these, as the temperature decreasing plot in Figure 6 seems to be too short for the data point to change due to the limit of accuracy.

Conclusion

In this lab, several features of LabView were examined using some electrical components. With LabView having advantages and disadvantages compared to regular code-based tools, it mostly features block programming with a clear user interface separately. The measured outcomes of temperature-responsive electric components basically meet all expectations. With the limit of measurements not visible at first glance in the graph in the front panel, a closer look to the results should be taken in the future to make corresponding adjustments. Only the boolean feature signaling certain events wasn't finished because time ran out.